

;Name: Tembhekar Sejal Popat

;Roll No:C02153

;Title: Write X86/64 ALP to convert 5- digit BCD number into its equivalent HEX number.

Section .data

Msg1 db "Enter 4 digit BCD number: " ; message asking user to enter number

Len1 equ \$-msg1 ; length of message

Msg2 db 10,"Equivalent HEX number: " ; message to display result (10 = newline)

Len2 equ \$-msg2 ; length of message

Section .bss

Num resb 5 ; buffer to store user input (4 digits + newline)

Hex resb 4 ; buffer to store 4-digit hex results

; macro for Linux system call

; parameters: syscall\_no , fd , buffer , length

%macro scall 4

Mov rax,%1

Mov rdi,%2

Mov rsi,%3

Mov rdx,%4

Syscall

%endmacro

Section .text

Global \_start

\_start:

;-----

; display message asking for input

;-----

```

Scall 1,1,msg1,len1    ; write msg1 to stdout

;-----

; read 4 digit BCD number from user

;-----

Scall 0,0,num,5        ; read input and store in num

;-----

; ASCII to decimal conversion
; Example: "1234" -> 1234

;-----

Mov rsi,num            ; point to input buffer

Xor rax,rax           ; clear rax (will store decimal value)

Mov rcx,4              ; process 4 digits

Dec_convert:

    Mov bl,[rsi]        ; load ASCII digit
    Sub bl,'0'          ; convert ASCII to numeric value
    Imul rax,10          ; multiply previous result by 10
    Add rax,rbx          ; add current digit
    Inc rsi             ; move to next digit
    Loop dec_convert     ; repeat 4 times

;-----

; Decimal to HEX conversion
; Example: 1234 -> 04D2

;-----

Mov rbx,16             ; divisor = 16 for hex conversion

Mov rdi,hex+3          ; start storing from last position

Mov rcx,4              ; generate 4 hex digits

```

Hex\_loop:

Xor rdx,rdx ; clear remainder register

Div rbx ; divide rax by 16

; quotient -> rax

; remainder -> rdx

Cmp dl,9 ; check if remainder > 9

Jbe digit ; if 0-9 skip adjustment

Add dl,7 ; convert 10-15 to A-F

Digit:

Add dl,'0' ; convert number to ASCII

Mov [rdi],dl ; store hex digit

Dec rdi ; move backward in buffer

Loop hex\_loop ; repeat for 4 digits

;-----

; display result

;-----

Scall 1,1,msg2,len2 ; print result message

Scall 1,1,hex,4 ; print 4-digit hex number

;-----

; exit program

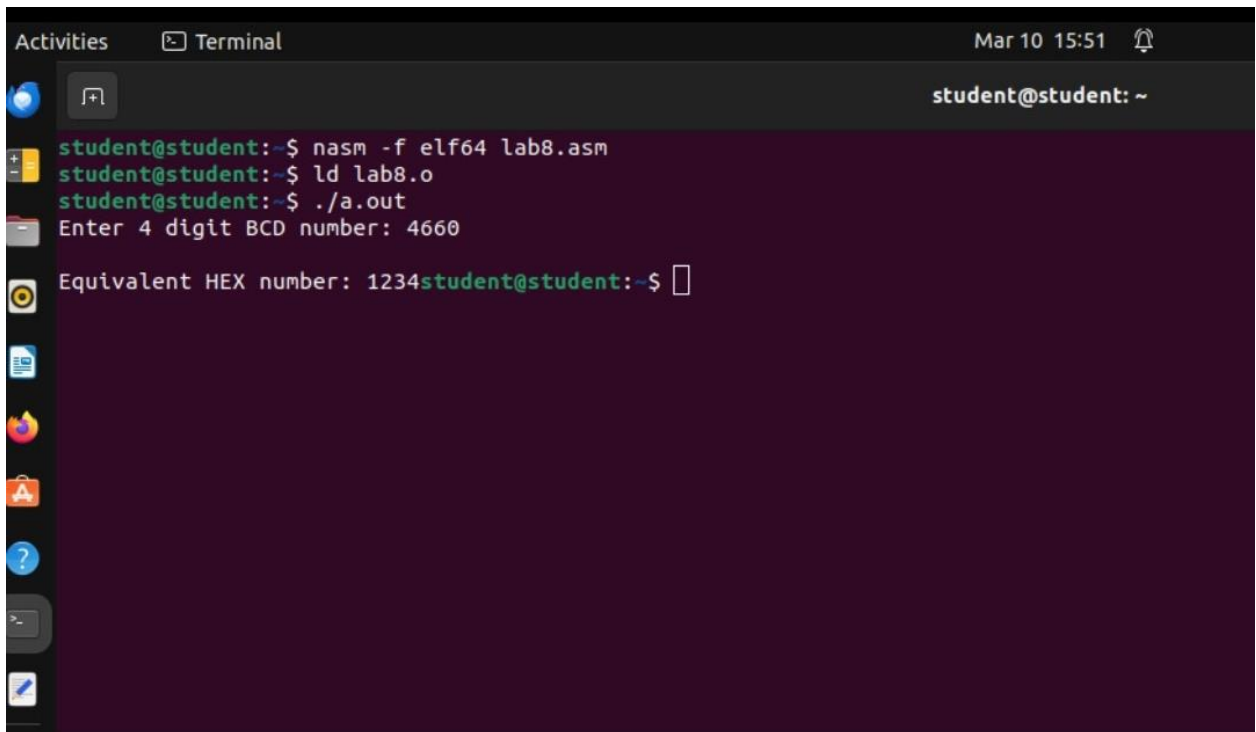
;-----

Mov rax,60 ; syscall number for exit

Xor rdi,rdi ; exit status 0

Syscall

Output :



A terminal window titled "Terminal" with a dark background. The window shows a series of commands and their outputs. The user is logged in as "student" on a machine named "student". The commands executed are: `nasm -f elf64 lab8.asm`, `ld lab8.o`, and `./a.out`. The output of the program prompts the user to "Enter 4 digit BCD number: 4660" and then displays "Equivalent HEX number: 1234". The terminal has a sidebar on the left with various application icons and a top bar showing the date and time as "Mar 10 15:51".

```
Activities Terminal Mar 10 15:51 student@student: ~
student@student:~$ nasm -f elf64 lab8.asm
student@student:~$ ld lab8.o
student@student:~$ ./a.out
Enter 4 digit BCD number: 4660
Equivalent HEX number: 1234student@student:~$
```